

Essentials in No Arbitrage and Probability

Part Two

Computational Methods in Finance, Code 21040

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Agenda I

- Financial Assets
- Investment Strategies, Values, Cashflows, Gains
- No Arbitrage and Market Completeness
- Valuing European options

The Market (discrete time) I

Securities:

- d risky securities with prices $S_i = \{S_i(t)\}_{t=0}^T$, for $i = 1, \dots, d$.

Return over $[t, t + \Delta t[$

$$\frac{S_i(t + \Delta t) - S_i(t)}{S_i(t)} \text{ is not known at } t$$

- The *locally riskless* security with price $B = \{B(t)\}_{t=0}^T$, whose price at time t is known at $t - \Delta t$. We also assume $B(0) = 1$.

The return

$$\frac{B(t + \Delta t) - B(t)}{B(t)} \text{ is known at } t$$

Moreover, we denote with $r(t)$ the locally riskless interest rate for period $[t, t + \Delta t[$

$$r(t) \Delta t = \frac{B(t + \Delta t) - B(t)}{B(t)}$$

The Market (discrete time) II

for $t = 0, \Delta t, \dots, T - \Delta t$.

The quantity $r(t)$ represents the net rate of return obtained by buying one unit of the security B at time t and reselling it at time $t + \Delta t$.

$r(t)$ is *known* at t .

To describe the evolution of the (risky) assets:

- Ω set of events (states of nature): finite in discrete time.

$$\Omega = \{\omega_1, \dots, \omega_K\}$$

- $\mathbb{P}$ probability measure on Ω , $\mathbb{P}(\omega_k) > 0$ for all ω_k .
(Investors know the probability distribution of the risky returns).

$\mathbb{P}$ -expected value of $S_3(t)$:

$$\mathbb{E}^{\mathbb{P}}[S_3(t)] = \sum_{\omega_k \in \Omega} S_3(t)(\omega_k) \mathbb{P}(\omega_k)$$

The Market (discrete time) III

- *Information structure, expectations, variables (See Part One)*

Relevant market processes I

[DEF] Dynamic Investment Strategies. A dynamic investment strategy ϑ is a $d + 1$ dimensional $\mathcal{P}$ -adapted process $\vartheta_j = \{\vartheta_j(t)\}_{t=0}^{T-\Delta t}$, with $j = 0, 1, \dots, d$.
 $\vartheta_j(t)$ = number of units of the j -th security at time t (and held on $[t, t + \Delta t]$) based on information $\mathcal{F}_t$.

[DEF] Value Process For a dynamic strategy ϑ , we call **value process** of ϑ , denoted by $V_\vartheta = \{V_\vartheta(t)\}_{t=0}^T$, the adapted stochastic process defined by

$$V_\vartheta(t) = \begin{cases} \vartheta_0(t) B(t) + \sum_{i=1}^d \vartheta_i(t) S_i(t), & t = 0, \dots, T - \Delta t \\ \vartheta_0(T - \Delta t) B(T) + \sum_{i=1}^d \vartheta_i(T - \Delta t) S_i(T), & t = T \end{cases} \quad (1)$$

Relevant market processes I

[DEF] Cashflow Process. For a dynamic strategy ϑ , we call **cashflow process** of ϑ , denoted by $C_{\vartheta} = \{C_{\vartheta}(t)\}_{t=0}^T$, the adapted stochastic process defined by

$$C_{\vartheta}(t) = \begin{cases} -V_{\vartheta}(0) & t = 0 \\ \vartheta_0(t - \Delta t) B(t) + \sum_{i=1}^d \vartheta_i(t - \Delta t) S_i(t) - V_{\vartheta}(t) & t = \Delta t, \dots, T - \Delta t \\ V_{\vartheta}(T) & t = T \end{cases} \quad (2)$$

Relevant market processes I

[DEF] Self-financing Dynamic Strategies. A dynamic strategy ϑ is said to be **self-financing** if

$$C_{\vartheta}(t) = 0, \quad t = \Delta t, \dots, T - \Delta t \quad (3)$$

By exploiting expression (2), we can immediately see that condition (3) can be rewritten as follows:

$$\underbrace{\vartheta_0(t - \Delta t) B(t) + \sum_{i=1}^d \vartheta_i(t - \Delta t) S_i(t)}_{\text{Value at } t \text{ of } \vartheta(t - \Delta t)} = V_{\vartheta}(t), \quad t = \Delta t, \dots, T - \Delta t$$

Of major interest in the valuation of European-type derivative securities.

Relevant market processes I

[DEF] Discounted Gain Process, multi-period case. For a strategy ϑ , we call **discounted gain process** of ϑ , denoted by $G_{\vartheta} = \{G_{\vartheta}(t)\}_{t=0}^T$, the adapted stochastic process defined by

$$G_{\vartheta}(t) = \begin{cases} \frac{V_{\vartheta}(t)}{B(t)} + \sum_{\tau=0}^t \frac{C_{\vartheta}(\tau)}{B(\tau)}, & t = 0, \dots, T - \Delta t \\ \sum_{\tau=0}^T \frac{C_{\vartheta}(\tau)}{B(\tau)}, & t = T \end{cases} \quad (4)$$

Multi-period Arbitrage Opportunities I

Violations of the Law of One Price. A multi-period financial market gives rise to violations of the law of one price if there exist two dynamic investment strategies ϑ and ϑ' generating the same cashflows in the future, i.e.

$C_{\vartheta}(t) = C_{\vartheta'}(t)$ for all $t = \Delta t, \dots, T$, but having different initial values, i.e. $V_{\vartheta}(0) \neq V_{\vartheta'}(0)$.

Arbitrage Opportunities of the 1st type. A multi-period financial market gives rise to arbitrage opportunities of the first type if there exists a dynamic strategy ϑ having the following properties:

$$V_{\vartheta}(0) \leq 0$$

$$C_{\vartheta}(t) \geq 0, \quad \text{for all } t = \Delta t, \dots, T$$

$$\mathbb{P}(C_{\vartheta}(\tau) > 0) > 0, \quad \text{for some } \tau \in \{\Delta t, \dots, T\}$$

Multi-period Arbitrage Opportunities I

Self financing arbitrage opportunities. A multi-period financial market gives rise to self financing arbitrage opportunities of the first type if there exists a dynamic strategy ϑ having the following properties:

$$V_{\vartheta}(0) \leq 0$$

$$C_{\vartheta}(t) = 0, \quad \text{for all } t = \Delta t, \dots, T - \Delta t$$

$$\mathbb{P}(V_{\vartheta}(T) \geq 0) = 1 \text{ and } \mathbb{P}(V_{\vartheta}(T) > 0) > 0$$

(In continuous time we will focus on self financing strategies)

Risk neutral probabilities and No Arbitrage. I

Notation: $\tilde{S}_j(t)$ is the present value of the time t price of security j , that is

$$\tilde{S}_j(t) = \frac{S_j(t)}{B(t)} \text{ for } t = 0, \dots, T$$

[DEF] Equivalent Martingale (or Risk Neutral Probability) Measure. An equivalent martingale measure (EMM) or a risk-neutral probability measure (RN) is

- a strictly positive probability $\mathbb{Q}$ on the possible scenarios at the final date T , i.e. $\mathbb{Q}(\omega_k) > 0$ for all $k = 1, \dots, K$,
- such that the present value of each risky security is a $\mathbb{Q}$ -martingale, i.e.

$$\tilde{S}_j(t) = \mathbb{E}^{\mathbb{Q}} \left[\tilde{S}_j(t + \Delta t) \middle| \mathcal{P}_t \right], \quad t = 0, \dots, T - \Delta t \quad (5)$$

for all $j = 1, \dots, d$.

Risk neutral probabilities and No Arbitrage. I

Equivalently:

① Eqn.(5) is equivalent to Eqn.(6)

$$S_j(t) = \mathbb{E}^{\mathbb{Q}} \left[\frac{S_j(t + \Delta t)}{1 + r(t)\Delta t} \middle| \mathcal{P}_t \right], \quad t = 0, \dots, T - \Delta t \quad (6)$$

for all $j = 1, \dots, d$.

② Eqn. (6) can be equivalently rewritten as:

$$\mathbb{E}^{\mathbb{Q}} \left[\frac{S_j(t + \Delta t) - S_j(t)}{S_j(t)} \middle| \mathcal{P}_t \right] = r(t)\Delta t, \quad t = 0, \dots, T - \Delta t$$

$\rightsquigarrow$ For this reason $\mathbb{Q}$ is called *risk neutral*.

Risk neutral probabilities and No Arbitrage. I

PROPOSITION: Given a probability, $\mathbb{Q}$, strictly positive on the possible scenarios at the final date T , the following statements are equivalent:

- 1 $\mathbb{Q}$ is an equivalent martingale measure;
- 2 the value and cashflow processes of every dynamic strategy ϑ satisfy the following condition:

$$V_{\vartheta}(t) = \begin{cases} E^{\mathbb{Q}} \left[\frac{V_{\vartheta}(t + \Delta t) + C_{\vartheta}(t + \Delta t)}{1 + r(t) \Delta t} \middle| \mathcal{P}_t \right], & t = 0, \dots, T - 2\Delta t \\ E^{\mathbb{Q}} \left[\frac{C_{\vartheta}(T)}{1 + r(T - \Delta t) \Delta t} \middle| \mathcal{P}_{T - \Delta t} \right], & t = T - \Delta t \end{cases} \quad (7)$$

- 3 the discounted gain process of every dynamic strategy ϑ is a martingale under $\mathbb{Q}$, that is:

$$G_{\vartheta}(t) = E^{\mathbb{Q}} [G_{\vartheta}(t + \Delta t) | \mathcal{P}_t], \quad t = 0, \dots, T - \Delta t, \quad (8)$$

Risk neutral probabilities and No Arbitrage. I

In particular, the previous Proposition yields for *self-financing strategies* ϑ



$$V_{\vartheta}(t) = \begin{cases} E^Q \left[\frac{V_{\vartheta}(t + \Delta t)}{1 + r(t) \Delta t} \middle| \mathcal{P}_t \right], & t = 0, \dots, T - 2\Delta t \\ E^Q \left[\frac{V_{\vartheta}(T)}{1 + r(T - \Delta t) \Delta t} \middle| \mathcal{P}_{T - \Delta t} \right], & t = T - \Delta t \end{cases}$$

- $\tilde{V}_{\vartheta}(t) = \frac{V_{\vartheta}(t)}{B(t)}$, **the discounted value of every self financing strategy**
 ϑ is a **martingale** under $\mathbb{Q}$, that is:

$$\tilde{V}_{\vartheta}(t) = E^Q \left[\tilde{V}_{\vartheta}(t + \Delta t) \middle| \mathcal{P}_t \right], \quad t = 0, \dots, T - \Delta t,$$

as for self-financing strategies $G_{\vartheta}(t) = \frac{V_{\vartheta}(t)}{B(t)} - V_{\vartheta}(0)$.

The First Fundamental Theorem of Asset Pricing I

Theorem 1FTAP. In a discrete time multi-period financial market the following statements are equivalent:

- 1 the market is arbitrage-free (NA);
- 2 there exists a risk-neutral probability measure $\mathbb{Q}$.

The First Fundamental Theorem of Asset Pricing I

Proof of 2. $\Rightarrow$ 1. Assume that there exists an equivalent martingale measure $\mathbb{Q}$. Suppose that a multi-period arbitrage opportunity ϑ can be found. Then

$$\sum_{t=0}^T \frac{C_{\vartheta}(t)}{B(t)} \geq 0 \text{ and } \mathbb{Q} \left[\sum_{t=0}^T \frac{C_{\vartheta}(t)}{B(t)} > 0 \right] > 0$$

and

$$E^{\mathbb{Q}} [G_{\vartheta}(T)] = E^{\mathbb{Q}} \left[\sum_{t=0}^T \frac{C_{\vartheta}(t)}{B(t)} \right] > 0, \quad (9)$$

because the equivalent martingale measure $\mathbb{Q}$ is strictly positive by definition. By the proposition in slide 25 the process G_{ϑ} is a martingale under $\mathbb{Q}$ for any ϑ , and by definition $G_{\vartheta}(0) = 0$, so that $E^{\mathbb{Q}} [G_{\vartheta}(T)] = G_{\vartheta}(0) = 0$.

Since such results contradicts (9), our thesis is proved.

The First Fundamental Theorem of Asset Pricing I

Proof of 1. $\Rightarrow$ 2

The result relies on the equivalence between multiperiod no-arbitrage and no-arbitrage for all one-period submarkets. In fact, no-arbitrage in all one-period submarkets is equivalent to the existence in all one period submarkets of a one-period risk neutral probability measure (by the 1st Fundamental Theorem of Asset Pricing in the one-period framework). For each one-period submarket, the existing one-period risk neutral probability defines the conditional (to the specific one-period submarket) risk neutral probability for the multiperiod market, as from definition (6).

Dynamic Completeness: the Notion I

[DEF] A **contingent claim** is any $\mathcal{P}$ -adapted process $X = \{X(t)\}_{t=\Delta t}^T$.
 $X(t)$ = the cashflow generated by the contingent claim X at time t .

[DEF] The contingent claim $X = \{X(t)\}_{t=\Delta t}^T$ is **attainable** with securities $B, S_1, \dots, S_d$ if there exists a dynamic investment strategy ϑ such that

$$C_{\vartheta}(t) = X(t), \quad t = \Delta t, \dots, T$$

[DEF] Dynamically Complete Multi-period Financial Market. We say that a discrete time multi-period financial market is dynamically complete if every contingent claim $X = \{X(t)\}_{t=\Delta t}^T$ is attainable.

The Second Fundamental Theorem of Asset Pricing I

Theorem 2FTAP. In a discrete time multi-period financial market the following statements are equivalent:

- 1 the market is both arbitrage-free and dynamically complete;
- 2 there exists one and only one equivalent martingale measure.

No arbitrage valuation of redundant securities I

- Initial market $\{B, S_1, \dots, S_d\}$: NA
- New security X is redundant, i.e. there exists $\vartheta^X = \{\vartheta^X(t)\}_{t=0}^{T-\Delta t}$ such that

$$C_{\vartheta^X}(t) = X(t)$$

for $t = \Delta t, \dots, T$.

- **The NA-VALUATION PROBLEM:**

Find the new security price process, $V_X = \{V_X(t)\}_{t=0}^T$, such that the extended market

$$\left\{ B, S_1, \dots, S_d, \underbrace{\{V_X, X\}}_{\text{new derivative}} \right\}$$

is still arbitrage-free.

No arbitrage valuation of redundant securities I

PRICING PROPOSITION. *If the new security is redundant, the following three conditions are equivalent:*

- 1 the extended market is arbitrage-free;
- 2 for every strategy ϑ^X replicating the new security, we have

$$V_X(t) = V_{\vartheta^X}(t) \quad (10)$$

for $t = 0, \dots, T$;

- 3 for every risk-neutral probability measure, $\mathbb{Q}$, of the initial market, we have

$$\begin{aligned} V_X(t) &= \mathbb{E}^{\mathbb{Q}} \left[\frac{X(t+\Delta t) + V_X(t+\Delta t)}{1+r(t)\Delta t} \middle| \mathcal{P}_t \right] \\ \text{for } t &= 0, \dots, T - 2\Delta t \text{ and} \\ V_X(T - \Delta t) &= \mathbb{E}^{\mathbb{Q}} \left[\frac{X(T)}{1+r(T-\Delta t)\Delta t} \middle| \mathcal{P}_{T-\Delta t} \right] \\ \text{for } t &= T - \Delta t. \end{aligned} \quad (11)$$

No arbitrage valuation of redundant securities I

Equation (11) can be written in the equivalent form

$$V_X(t) = \mathbb{E}^{\mathbb{Q}} \left[\sum_{\tau=t+\Delta t}^T \frac{B(t)}{B(\tau)} X(\tau) \middle| \mathcal{P}_t \right], \quad (12)$$

for $t = 0, \dots, T - \Delta t$. Equation (12) says that the time- t price of the security X is equal to the $\mathcal{P}_t$ -conditional risk neutral expected value of the future cashflows discounted at time t , as

$$\frac{B(\tau)}{B(t)} = \prod_{s=t}^{\tau-\Delta t} (1 + r(s)\Delta t) \text{ is the discount factor on } [t; \tau[$$

Valuation and replication of European Options I

If the derivative is European derivative with maturity T , no intermediate cashflows

$$X(t) = 0 \text{ for } t = \Delta t, \dots, T - \Delta t.$$

$\Rightarrow$ The replicating strategy ϑ (if any) is self-financing strategy.

The no-arbitrage price $V_X(t)$ of X and the value of the replicating strategy coincide and are given by

$$\frac{V_X(t)}{B(t)} = \frac{V_\vartheta(t)}{B(t)} = \mathbb{E}^{\mathbb{Q}} \left[\frac{X(T)}{B(T)} \middle| \mathcal{P}_t \right]$$

for any $t = 0, \dots, T$ (Pricing proposition).

Valuation and replication of European Options I

In order to compute $V_X(0)$ we can either compute the expectation of $\frac{X(T)}{B(T)}$ under $\mathbb{Q}$, or exploit the following *backward recursion in time*:

$$\frac{V_X(T)}{B(T)} = \frac{X(T)}{B(T)}$$

and for $t = T - \Delta t, \dots, 0$

$$\frac{V_X(t)}{B(t)} = \mathbb{E}^{\mathbb{Q}} \left[\frac{V_X(t + \Delta t)}{B(t + \Delta t)} \middle| \mathcal{P}_t \right]$$

The backward recursion exploits (and is equivalent to) the $\mathbb{Q}$ -martingality of the discounted value of the European derivative X . *Starting from the terminal value* $V_X(T) = X(T)$, we determine by backward induction $V_X(t)$ from the value of

Valuation and replication of European Options II

V_X at the step before, i.e. at $t + \Delta t$, for $t = T - \Delta t, \dots, 0$. This approach is precious when dealing with American options, because it can be generalized to account for the early exercise premium.